

Seungmin Han

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Professional Summary

Actuarial consultant with 8+ years of experience in life and P&C insurance, specialising in experience studies, IFRS 17, K-ICS, and structured reinsurance solutions. Proven track record of developing and maintaining actuarial models for valuation, pricing, and regulatory reporting at major Korean insurers, with direct exposure to APAC reinsurance markets through Samsung Re's Singapore core system project. Combines deep Korean insurance market knowledge with Python/SQL-driven data analytics and AI-powered process automation to deliver competitive, risk-adjusted reinsurance terms and innovative structured solutions.

Experience

PwC Consulting | Manager

Seoul, Korea | 2022.04 – Present

- Designed and executed K-ICS ratio enhancement strategies for Korean life insurers, including a 5-year capital management roadmap incorporating dynamic ALM and optimised investment portfolio allocation, directly improving regulatory capital ratios and P&L stability under rising interest rate environments.
- Architected core system upgrade requirements for Samsung Re's Singapore subsidiary, translating complex, multi-stakeholder business needs from Korean and international counterparts into a unified technical strategy — providing direct exposure to reinsurance treaty structures and APAC market dynamics.
- Led AI-powered transformation of P&C risk engineering and claims workflow at Samsung Fire & Marine, adapting an LLM to fully automate key actuarial processes and enhance operational efficiency, with Python-based data pipelines underpinning the automation.
- Designed a comprehensive risk monitoring dashboard for Hana Securities, defining KRIs across market, credit, and liquidity risk dimensions to provide C-level management with real-time consolidated views for strategic decision-making.

SiG Partners | Senior Consultant

Seoul, Korea | 2017.05 – 2022.03

- Specialised in full-lifecycle IFRS 17 closing system and actuarial assumption system implementation for multiple Korean life insurers, developing deep expertise in actuarial model architecture, liability data flows, and assumption governance frameworks.
 - Owned end-to-end implementation of Shinhan Life's IFRS 17 closing system and actuarial assumption platform (2017–2021), covering model development, data validation, quarterly reporting processes, and stakeholder sign-off.
 - Conducted IFRS 17 liability closing system audit for Kyobo Life Insurance, verifying regulatory compliance and data accuracy across valuation and reporting pipelines.
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Project Experience — Actuarial & Reinsurance Highlights

Experience Studies & Valuation

Shinhan Life Insurance | IFRS17 Closing System Implementation | 2017.07 – 2021.06 Owned the full implementation lifecycle for IFRS 17 closing and actuarial assumption systems. Developed and maintained actuarial models, prepared and validated model input data, and led quarterly reporting processes including valuation, monitoring, and continuous improvement initiatives — directly analogous to Munich Re's experience studies and quarterly reporting requirements.

Kyobo Life Insurance | IFRS17 & K-ICS Audit | 2021.11 – 2022.03 Conducted comprehensive audit of IFRS 17 liability closing system, verifying regulatory compliance and ensuring data accuracy across valuation pipelines.

Structured Solutions & Capital Management

IBK Pension Insurance | K-ICS Ratio Enhancement Strategy | 2023.09 – 2023.12 Led development of a 5-year strategic risk management roadmap to enhance financial stability, incorporating dynamic ALM strategies and a fully optimised investment portfolio — directly supporting capital, tax, and volatility considerations central to structured reinsurance solutions.

NongHyup Life Insurance | ALM Strategy Formulation | 2023.04 – 2023.07 Architected a 5-year ALM strategy that successfully mitigated P&L impact of rising interest rates, demonstrating ability to structure transactions that address client-specific capital and volatility considerations.

APAC Reinsurance Market Exposure

Samsung Re | Core System Upgrade (Singapore Subsidiary) | 2024.01 – 2024.05 Led architectural design of core system upgrade for Samsung Re's Singapore operations. Bridged communication between Korean HQ and international stakeholders across APAC, forging a single aligned technical strategy — providing direct exposure to reinsurance market dynamics and cross-border regulatory frameworks.

AI-Driven Analytics & Automation

Samsung Fire & Marine | AI-Powered Core System Transformation | 2024.05 – Present Leveraged P&C insurance domain expertise and AI to transform risk engineering and claims workflows using LLM-based automation, with Python-driven data pipelines for model input preparation and validation.

Hana Securities | Risk Monitoring Dashboard | 2025.10 – 2025.11 Designed and implemented a real-time risk monitoring dashboard defining KRIs across market, credit, and liquidity risk for C-level strategic decision-making.

Education

Pennsylvania State University, University Park, PA Bachelor of Science in Risk Management (Actuarial Science Option) | 2010.08 – 2017.04 | GPA: 3.⁴⁸/_{4.0}

Certifications & Professional Development

- **Associate of the Society of Actuaries (ASA):** In progress, final exam pending
 - **Chartered Enterprise Risk Analyst (CERA):** Actively pursuing designation
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Technical Skills

Category	Skills
Actuarial & Risk	IFRS 17, K-ICS, ALM, Experience Studies, Valuation, Solvency II, Capital Management, Pricing
Programming & Data	Python (data manipulation, analysis, automation), SQL, data pipeline development
AI & Technology	LLM Implementation, AI Process Automation, System Architecture
Reinsurance	Structured Solutions, Treaty Review, Profitability Analysis, APAC Market Exposure
Languages	Korean (Native), English (Working Proficiency — verbal and written)